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Explore factors affecting students' acceptance of AI resurrection and ethical dilemmas in AI resurrection technology among Chinese college students: a cross-sectional study

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Abstract

Background Advances in artificial intelligence (AI) have brought AI resurrection, the digital reconstruction of the deceased closer to reality. Understanding the psychological factors associated with acceptance of this technology is essential for informing its ethical and educational implications.

Methods This cross-sectional study investigated Chinese college students' acceptance of AI resurrection and examined demographic, death-related background factors, and AI-related predictors. A total of 990 valid responses were collected through an online self-administered questionnaire (Cronbach's $\alpha = 0.799$). Spearman's correlation and multinomial logistic regression analyses were conducted to examine associations between variables and acceptance levels across five hypothetical scenarios.

Results Higher AI engagement and stronger perceived harm predictability were positively associated with acceptance of AI resurrection technologies, whereas fear of AI demonstrated scenario-dependent effects. Participants who had experienced the death of loved ones or studied medicine showed greater acceptance. Nearly half of the respondents (47.9%) believed that both the deceased's and family's wishes should be considered in resurrection decisions.

Conclusions Cognitive and affective factors were associated with individuals' acceptance of AI resurrection technologies. These findings contribute to understanding human-AI interaction in existential contexts and emphasize the importance of consent and data protection in future applications.

Keywords AI resurrection, Psychological acceptance, AI fear, Harm predictability, Human-technology interaction, Death-related attitudes, Chinese college students

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Background

As artificial intelligence (AI) increasingly intersects with fundamental aspects of human life and death, understanding its psychological and ethical implications has become a critical concern. AI has rapidly advanced to a stage where technologies can replicate human voices, images, and personalities from digital traces, enabling what has recently been termed AI resurrection. This concept refers to the digital reconstruction of a deceased individual through AI systems capable of generating interactive simulations or chatbots [1, 2]. While once confined to speculative fiction, the growing capabilities of generative AI systems such as ChatGPT and DeepSeek have transformed these ideas into feasible technological possibilities [3, 4]. These developments raise profound psychological questions about how humans respond to technologies that blur the boundary between the living and the dead. Existing studies have examined the ethical and legal implications of posthumous data use and digital personhood [5–7]. However, empirical research on individuals' psychological acceptance of AI resurrection remains scarce. Understanding public perceptions of such technologies is essential, as social acceptance strongly predicts adoption and compliance behaviours toward emerging AI systems [8, 9].

University students represent a particularly relevant demographic for investigating these issues. They are early adopters of digital technology, frequently engage with AI in educational and personal contexts, and will likely shape the ethical and societal norms surrounding future AI use [10, 11]. Moreover, cultural beliefs about death and continuity of the self, particularly those informed by Confucian and Daoist traditions, may inform emotional and moral evaluations of digital resurrection technologies [2, 12]. These cultural frameworks emphasize relationality, reverence for ancestors, and the moral significance of remembrance, providing a cultural lens through which participants may interpret AI-mediated representations of the dead. Accordingly, the discussion of cultural traditions in this study reflects an interpretive context rather than direct statistical testing.

Unlike conventional AI applications that primarily support efficiency or decision-making, AI resurrection involves deeply personal and existential domains, including mortality, identity continuity, and posthumous autonomy. These characteristics raise questions about whether traditional technology acceptance frameworks remain applicable in such contexts. Conceptually, even in existentially sensitive contexts, judgments of acceptability involve cognitive evaluation and emotional response to an AI system.

From this perspective, the Technology Acceptance Model (TAM) [8] and the Human–AI Trust Framework [13] remain theoretically relevant, as they offer

complementary lenses for understanding how individuals assess, trust, and emotionally react to AI technologies in morally charged contexts. TAM provides a foundation for examining how cognitive evaluations, such as familiarity, perceived ease of interaction, and anticipated usefulness, shape acceptance. In the context of AI resurrection, these evaluations extend beyond instrumental utility to include judgments about psychological comfort and perceived appropriateness of interacting with a posthumous digital entity.

At the same time, the Human–AI Trust Framework is particularly pertinent to AI resurrection because trust-related beliefs, such as predictability, transparency, and controllability, become central when AI systems operate in ethically sensitive domains. When AI technologies engage with death and identity, uncertainty regarding potential harm or misuse may undermine trust and amplify emotional resistance.

Therefore, integrating TAM with trust-based perspectives allows for a more nuanced understanding of how cognitive familiarity, perceived predictability, and affective responses jointly shape acceptance of AI resurrection. These frameworks are employed as psychologically grounded models for examining how complex moral concerns are translated into acceptance judgments. By situating these frameworks within the Chinese socio-cultural context and considering death-related beliefs and AI-related affect (e.g., AI fear), this study aims to extend our understanding of how AI resurrection might be evaluated within a broader cultural and moral context.

The psychological acceptance of AI technologies

Building on the above theoretical positioning, psychological acceptance of technology has been extensively examined through the Technology Acceptance Model (TAM), which posits that behavioural intention is determined by cognitive evaluations such as perceived usefulness and perceived ease of use [8]. TAM and its extensions, such as the Unified Theory of Acceptance and Use of Technology [9], have been applied across various contexts, including healthcare [14], education [15], and social media [16].

Within this literature, TAM primarily conceptualizes acceptance as a cognitive evaluation process, emphasizing individuals' assessments of usefulness, ease of interaction, and familiarity with technology. Traditional TAM frameworks mainly emphasize cognitive rather than emotional processes. Subsequent research suggests that affective factors, including trust, anxiety, and fear, play critical roles in shaping user attitudes toward AI [13, 17]. In particular, trust in AI depends on users' perceptions of predictability, transparency, and controllability of AI behaviour [18, 19]. Conversely, AI-related fear or anxiety may be associated with heightened perceived risks and moral unease and lower acceptance [20, 21].

Accordingly, integrating cognitive and affective dimensions has become increasingly important for understanding acceptance of AI systems that involve uncertainty and ethical sensitivity.

The role of cognitive engagement and harm predictability

AI engagement, defined as the degree of familiarity, understanding, and frequency of use of AI tools, reflects the cognitive component of interaction. Higher engagement enhances perceived ease of use and reduces uncertainty, which is associated with higher acceptance [14, 21]. Research in educational psychology also shows that repeated exposure to AI systems increases self-efficacy and technology readiness [22, 23].

A related construct, AI harm predictability, refers to users' belief that AI behaviour is transparent and controllable, which are concepts central to the Human-AI Trust Framework [13]. When people perceive AI systems as predictable and aligned with human values, they are more likely to trust and accept them [13, 18, 21]. Transparency in AI decision-making can mitigate uncertainty and enhance psychological safety [24, 25]. These findings suggest that cognitive variables, namely engagement and perceived predictability, may serve as key facilitators of acceptance toward ethically complex AI applications.

The role of AI fear

In contrast, emotional responses such as AI fear can inhibit acceptance. Fear of AI encompasses apprehension about potential harm, ethical misuse, or loss of human control [20]. This affective response is often rooted in perceived threats to identity, autonomy, or existential security [26, 27]. From a psychological perspective, fear may be linked to avoidance tendencies and moral resistance, and has been associated with lower acceptance of technologies perceived as invasive or dehumanizing. Studies have shown that individuals experiencing higher AI-related fear are less likely to adopt AI-driven systems regardless of their functionality [28, 29]. In contexts involving mortality or posthumous data, fear may intensify because of perceived violations of human dignity and spiritual boundaries [2, 30]. Therefore, examining AI fear alongside cognitive variables is crucial for understanding affective resistance toward AI resurrection.

Taken together, this study conceptualizes cognitive and affective factors as parallel yet interacting pathways influencing acceptance of AI resurrection. Cognitive evaluations, such as AI engagement and perceived harm predictability, reflect individuals' rational assessments of familiarity, controllability, and trustworthiness of AI systems. In parallel, affective responses, particularly AI-related fear, capture emotional reactions rooted in ethical concern, existential unease, and perceived moral risk.

Rather than assuming a fixed hierarchical relationship between cognition and emotion, this study adopts a context-sensitive perspective in which affective responses operate as a parallel process whose influence varies across contexts. Specifically, AI fear is not treated as a simple antecedent to trust-related beliefs, but as an affective process whose influence varies depending on consent conditions. When consent is unclear or absent, fear may heighten moral resistance and suppress acceptance. Conversely, when explicit consent is present, fear may be reinterpreted or attenuated, resulting in different patterns of association with acceptance.

This conceptualization aligns with dual-process models of decision-making and human-AI interaction, which suggest that cognitive and emotional processes jointly, though not uniformly, shape judgments under varying contextual conditions. Accordingly, the present study examines how cognitive trust-related factors and affective responses operate simultaneously, while allowing their relative influence to differ across consent-based scenarios.

The present study

Building on these theoretical and contextual foundations, the present study aims to explore the psychological and contextual factors that shape college students' acceptance of AI resurrection technologies in China. Accordingly, the Chinese cultural context is considered as an interpretive background that informs participants' meaning-making processes, rather than as a formally operationalized predictor in the analytical models.

Instead of testing predefined hypotheses, this research adopts an exploratory analytic approach guided by the Technology Acceptance Model [8] and the Human-AI Trust Framework [13].

Specifically, we examine how cognitive factors (AI engagement and AI harm predictability), affective factors (AI fear), and contextual or personal characteristics (demographics and death-related experiences) relate to individuals' acceptance of AI resurrection across multiple hypothetical scenarios that differ in consent conditions [30].

Given the exploratory nature of this study and the ethical sensitivity of AI resurrection, we did not assume that the same psychological mechanisms would operate uniformly across all consent scenarios. Instead, the scenario-based analytical strategy was designed to examine whether cognitive and affective predictors of acceptance remain consistent or vary depending on consent context.

By employing multinomial logistic regression models across multiple scenarios, this study explicitly allows for the possibility that different consent conditions (e.g., unclear consent, parental consent, or self-consent) may activate distinct psychological processes. This approach

enhances transparency by aligning the research questions with the analytical strategy and enables a more nuanced interpretation of scenario-dependent findings.

Accordingly, the study addresses the following research questions:

1. Which demographic and death-related factors are associated with acceptance of AI resurrection across consent scenarios?
2. How do cognitive (trust-related) and affective (fear-related) dimensions influence acceptance when considered simultaneously?
3. Do these associations differ across consent-based contextual scenarios (e.g., unknown consent, parental consent, or self-consent)?

This is the first empirical study to examine AI resurrection acceptance in a Chinese cultural context. By focusing on young adults in a rapidly digitizing society, this research provides empirical evidence for understanding the psychological mechanisms underlying public acceptance of AI technologies that intersect with death, autonomy, and identity.

Method

Study design and setting

This study aimed to investigate the psychological and contextual factors influencing Chinese college students' acceptance of AI resurrection technologies across five scenarios. This study employed a cross-sectional design to collect data from Chinese college students. Convenience sampling was used to recruit participants from seven geographical regions across China, covering East, North, Central, South, Southwest, Northwest, and Northeast China, to improve geographic diversity. Participants were recruited through online distribution of the questionnaire link via class-based instant messaging groups across institutions in different regions, with the assistance of student gatekeepers. Eligible participants were current associate, undergraduate, master's, or doctoral students.

The sample was relatively homogeneous in terms of age, with 96% of participants concentrated in the 17–29 years range, and over half were medical students. These factors should be considered when interpreting the generalizability of the findings, as the sample may not fully reflect the broader Chinese population. While these characteristics align with the theoretical focus of the study on young, highly educated individuals, the results should not be overgeneralized to the wider population.

Data were collected through a self-designed online questionnaire distributed via “Wenjuanxing,” a one-stop survey research platform.

In particular, the relatively high proportion of medical students is of theoretical relevance, as medical education often involves repeated exposure to mortality, end-of-life decision-making, and ethical reflection. These factors are likely to influence attitudes toward both AI technologies and end-of-life care. Thus, academic major was treated as a key analytical variable, enabling us to examine whether exposure to medical education is associated with differential acceptance of AI resurrection technologies.

To ensure an adequate final sample after data quality screening, a total of 1002 questionnaires were distributed. After excluding ineligible or invalid responses, 990 valid cases were retained for analysis, yielding an effective response rate of 98.8%.

Sample adequacy was evaluated based on contemporary statistical considerations rather than simple rules of thumb. Given the large sample size ($N = 990$) and the use of scenario-specific multinomial logistic regression models, the study meets commonly recommended criteria for model stability and reliable parameter estimation, including sufficient observations per outcome category and per estimated parameter. Methodological guidelines for logistic and multinomial logistic regression commonly recommend at least 10–20 observations per estimated parameter within each outcome category to ensure stable model estimation [31, 32]. Accordingly, the current sample size is considered sufficient to support reliable estimation and interpretation of scenario-specific effects.

Prior to participation, respondents read an informed consent statement describing the study's purpose, voluntary nature, confidentiality, and right to withdraw at any time. Duplicate submissions and responses completed in under two minutes were excluded to ensure data quality. On average, participants took 10–15 min to complete the survey.

The survey comprised five sections: [1] demographic information [2], death-related beliefs and experiences [3], knowledge and attitudes toward AI [4], five hypothetical AI resurrection scenarios and related questions, and [5] an ethical question on whether the rights of the deceased should be considered in AI resurrection.

While the study uses single-item measures for AI harm predictability and AI fear, we acknowledge that single-item measures have limitations in terms of construct validity and reliability. These measures were chosen for their simplicity and efficiency in an exploratory study with a large sample size. However, we recognize that single-item measures may capture a limited scope of the construct and could be subject to measurement error. Future studies should aim to use multi-item validated scales to provide more reliable and comprehensive assessments of AI-related fears and harm predictability.

The full instrument is provided in the Appendix.

This study was conducted in accordance with the ethical principles of the Declaration of Helsinki and was approved by the Ethics Committee of Xiangya School of Nursing, Central South University. All data were anonymized and securely stored in encrypted files accessible only to the research team.

Survey measures

The questionnaire was designed based on the Technology Acceptance Model (TAM) [8] and the Human–AI Trust Framework [13], incorporating vignette-based adaptations from Iwasaki [30] to capture ethical and emotional responses. It consisted of five major sections: [1] sociodemographic characteristics [2], death-related beliefs and experiences [3], AI engagement [4], AI harm predictability, and [5] Acceptance of AI resurrection across five hypothetical scenarios.

The questionnaire was pretested with five graduate students and refined for clarity and reliability. The final version achieved a Cronbach's α of 0.799. All items used a 5-point Likert scale unless otherwise stated. The full wording of the items and response options is available in the Appendix.

Sociodemographic characteristics

Participants reported gender, age, educational level (associate, undergraduate, master's, or doctoral degree), and academic major. Majors were subsequently grouped into medical and non-medical disciplines for analysis, as prior studies have shown that medical students often display greater exposure to ethical dilemmas, death-related experiences, and technology-assisted care, which may shape their attitudes toward both AI and end-of-life technologies [10].

Death-related beliefs and experiences

Two closed-ended items measured participants' death-related beliefs and personal experiences related to death. The first item assessed afterlife belief: "Do you believe death represents the end of life, or do you believe in an afterlife or supernatural continuation (e.g., soul or spirit)?" (response options: "End of life" or "Afterlife/spirit exists"). Previous research has shown that belief in a soul or afterlife is negatively associated with moral approval of digital immortality technologies such as mind-uploading [33]. The second item asked whether participants had experienced the loss of a close family member or friend ("Yes" or "No"). Prior work with bereaved populations demonstrates that such experiences influence grief-related digital behaviours and attitudes toward posthumous communication technologies [34]. These two items capture participants' death-related belief orientation and bereavement experience, which may shape their moral evaluation of AI resurrection technologies.

AI engagement

Grounded in the Perceived Ease of Use (PEOU) dimension of the Technology Acceptance Model [8], AI engagement captured participants' cognitive familiarity and experiential interaction with AI systems, which are known to facilitate perceived ease and reduce technological uncertainty [14, 21]. Participants indicated [1] how frequently they used AI technologies (e.g., chatbots, generative AI tools, or voice assistants) and [2] how well they understood these technologies, both measured on a 5-point Likert scale (1 = "Not at all" to 5 = "Very much"). The mean of the two items represented participants' level of AI engagement, with higher scores reflecting greater familiarity and perceived ease of interaction.

AI harm predictability

Based on the Human–AI Trust Framework [13], this construct represents a cognitive antecedent of trust, capturing perceived controllability and transparency of AI behaviour. Participants were asked, "To what extent do you believe the potential harm caused by AI technologies can be predicted and managed?" Responses were rated on a 5-point Likert scale (1 = "Not at all predictable" to 5 = "Highly predictable"). Higher scores indicated stronger beliefs that AI operates in a controllable and transparent manner, aligned with human values and social norms.

Perceived predictability has been shown to enhance users' cognitive trust and willingness to accept AI-mediated systems [13, 18, 35].

AI fear

Drawing on the affective extensions of the Technology Acceptance Model [9, 21] and the Human–AI Trust Framework [13], AI fear represents an affective concern about the potential misuse and ethical risks of AI technologies, reflecting emotional unease toward AI in morally sensitive contexts.

Participants rated their agreement with the statement: "I am concerned that AI technologies, including AI resurrection, could be misused or cause ethical problems." Responses were recorded on a 5-point Likert scale (1 = "Strongly disagree" to 5 = "Strongly agree").

Higher scores indicate stronger emotional unease, reflecting fear of ethical misuse, loss of control, or identity threat associated with AI systems. Prior studies have shown that such affective responses negatively predict acceptance and trust in AI-driven technologies [20, 21, 26].

AI resurrection scenarios and acceptance

To examine social and personal acceptance of AI resurrection, five hypothetical scenarios were adapted from Iwasaki [30]. Each scenario described a case in which a young woman, Xiaotong, was digitally resurrected using

AI technology under varying consent conditions (e.g., unknown wishes, parental consent, self-consent). Participants rated the social acceptability of each case on a 5-point Likert scale (1 = “Unacceptable” to 5 = “Acceptable”). Two additional vignettes asked participants to imagine themselves as (a) the deceased and (b) the deceased’s parent, rating their agreement with statements regarding their own or their child’s resurrection. The final question assessed perceived ethical rights, asking: “Whose wishes should be prioritized in AI resurrection? the deceased, the family, or both?”

Data analysis

All statistical analyses were conducted using SPSS version 27.0 (IBM Corp., Armonk, NY, USA). Statistical significance was considered at $p < .05$ (two-tailed). A two-stage analytical strategy was employed.

First, descriptive statistics were computed for all variables, expressed as mean (SD) for continuous variables and as frequencies and percentages for categorical variables.

Second, Spearman’s rank correlation analyses were performed to examine bivariate associations among demographic, death-related, and AI-related variables and participants’ acceptance of AI resurrection across the five

hypothetical scenarios. Variables that showed significant correlations ($p < .05$) were subsequently included in multinomial logistic regression models to identify predictors of AI resurrection acceptance in each scenario.

For demographic factors, three independent variables were analysed: gender, education, and academic major. Gender was included as a binary categorical variable. Academic major was dichotomized into medical and non-medical fields. Educational level was grouped into three categories (associate, undergraduate, and postgraduate).

For death-related beliefs and experiences, two dichotomous variables were included: [1] belief orientation (“end of life” vs. “afterlife/spirit exists”) and [2] prior experience of losing a close person (“yes” vs. “no”).

For AI-related attitudes, three continuous variables were analysed: AI engagement, AI harm predictability, and AI fear. In the regression models, continuous variables were categorized into three levels to facilitate interpretation: low ([1, 2]), moderate [3], and high ([4, 5]). Education level was also grouped into three categories (associate, undergraduate, postgraduate).

This categorization approach is consistent with prior exploratory research on ethically sensitive and posthumous technologies, such as Iwasaki’s study on digital cloning, and was adopted to facilitate scenario-based interpretation, with results interpreted as indicative rather than exact effect estimates [30].

Table 1 Demographic characteristics of participants (N = 990)

	N	%
Gender		
Woman	658	66.5
Man	332	33.5
Age		
17–29	974	96.0
30 or more	16	4.0
Education		
Associate degree	89	9.0
Bachelor’s degree	738	74.5
Master’s degree	117	11.8
Doctor’s degree	46	4.6
Major		
Philosophy	156	15.8
Economics	20	2.0
Law	30	3.8
Medicine	508	51.3
Education	14	1.4
Agriculture	7	0.7
Engineering	114	11.5
Science	68	6.9
Management	25	2.5
Literature	24	2.4
Military Science	1	0.1
History	4	0.4
Art	11	1.1

N = 990 Participants were on average 21.2 years old (SD = 3.4), SD Standard Deviation

Result

A total of 1002 participants were approached, of whom 12 were excluded due to ineligibility. The final sample comprised 990 valid responses, yielding an effective response rate of 98.8%. Descriptive, correlational, and regression analyses were performed to examine associations between demographic, cognitive, and affective variables and acceptance of AI resurrection.

Participant characteristics

Participants’ demographic characteristics are summarized in (Table 1). The majority were female (66.5%), and most were aged between 17 and 29 years (96.0%). The largest proportion was undergraduate students (74.5%), and medical majors accounted for 51.3% of the total sample.

Death-related beliefs and experiences

Table 2 summarizes participants’ death-related beliefs and experiences. Most respondents (80.6%) believed that death represents the end of life, whereas 19.4% endorsed beliefs in an afterlife or spiritual continuation. In addition, 76.6% reported having experienced the loss of a close family member or friend.

Table 2 Response distribution of death-related cultural background ($N=990$)

	<i>N</i>	%
What is the view of death in your cultural background?		
Death is the end of life	798	80.6
Others (afterlife or the existence of the soul)	192	19.4
Have you ever suffered the loss of a loved one (including but not limited to family members, relatives, or friends) and felt grief or regret?		
Yes	758	76.6
No	232	23.4

$N=990$. Values are presented as frequency (%)

Table 3 Descriptive statistics for AI-related variables ($N=990$)

	Mean	SD
AI engagement	3.085	0.943
AI harm predictability	3.050	1.075
AI fear level	3.930	0.904

$N=990$, SD Standard Deviation, Higher mean scores indicate greater levels of the measured construct

Table 4 Descriptive statistics for responses on societal and personal perspectives ($N=990$) Panel A: Mean, SD

	Mean	SD
Societal: consent unclear	2.88	1.018
Societal: with parental consent	3.26	1.033
Societal: with the deceased's consent	3.40	1.037
Personal: self	2.96	1.211
Personal: as parents	2.88	1.158

Panel B: response distribution (%)

AI-related variables

Descriptive statistics for AI-related variables are presented in (Table 3). Three independent variables were measured: AI engagement, AI harm predictability, and AI fear, each rated on a 5-point Likert scale. AI engagement was assessed using two items adapted from Iwasaki [30]. AI engagement and AI harm predictability showed mean values around the midpoint of the scale. In contrast, AI fear exhibited a higher mean score, approaching the upper range of the scale.

Table 5 Response distribution regarding ethical consideration of the deceased's rights ($N=990$)

	<i>N</i>	%
The deceased's pre-mortem wishes should be fully respected.	435	43.9
Balance the wishes of the deceased and the family.	474	47.9
Decide based on the family's wishes.	62	6.3
Other (please specify): _____	19	1.9

$N=990$. Values are presented as frequency (%)

Acceptance of AI resurrection

Table 4 summarizes participants' acceptance of AI resurrection across different consent conditions. Acceptance was lowest when the deceased's consent was unknown and increased in scenarios involving explicit consent, such as parental or self-consent.

Panel B: response distribution (%)

	Unacceptable	Neutral	Acceptable
Societal: consent unclear	30.9	43.6	25.5
Societal: with parental consent	18.5	38.9	42.6
Societal: with the deceased's consent	14.9	37.1	48.0
	Disagree	Neutral	Agree
Personal: self	30.9	35.4	33.7
Personal: as parents	33.1	36.4	30.5

Abbreviations: $N=990$, SD Standard Deviation, Panel A shows mean scores (1–5 Likert scale), Panel B presents frequency percentages for each response category.

Table 5 presents participants' views on whose wishes should take ethical priority in AI resurrection.

Most respondents emphasized either prioritizing the deceased's wishes or balancing the wishes of the deceased and the family.

Correlation analysis

Table 6 presents Spearman's rank correlations between demographic, death-related, and AI-related variables and acceptance of AI resurrection across five scenarios. AI engagement and AI harm predictability were positively

Table 6 Spearman correlation analysis across five AI resurrection scenarios ($N=990$)

Scenario	1Societal: consent unclear		2Societal: with parental consent		3Societal: with the deceased's consent		4Personal: self		5Personal: as parents	
	<i>r</i>	<i>p</i>	<i>r</i>	<i>p</i>	<i>r</i>	<i>p</i>	<i>r</i>	<i>p</i>	<i>r</i>	<i>p</i>
Gender	−0.016	0.624	−0.014	0.665	0.044	0.166	−0.026	0.409	−0.050	0.118
Education	−0.057	0.074	0.007	0.832	0.043	0.176	−0.075	0.019*	−0.034	0.287
Major	0.067	0.036*	0.010	0.745	0.001	0.963	0.066	0.037*	0.028	0.383
AI engagement	0.107	0.001*	0.218	0.000*	0.206	0.000*	0.141	0.000*	0.155	0.000*
AI harm predictability	0.419	0.000*	0.355	0.000*	0.330	0.000*	0.400	0.000*	0.416	0.000*
AI fear level	−0.145	0.000*	0.156	0.000*	0.193	0.000*	−0.033	0.304	−0.071	0.026*
Belief	0.051	0.111	0.070	0.027*	0.026	0.412	0.031	0.331	0.009	0.768
Experience	0.074	0.020*	0.054	0.091	−0.004	0.891	0.028	0.374	0.018	0.565

Abbreviations: $N=990$, *r* Spearman's correlation coefficient, *p* significance level, $p < .05$ indicates statistical significance

Table 7 Multinomial logistic regression predicting acceptance of AI resurrection under unclear consent ($N=990$)

	β	SE	Wald	(P)	OR	95% CI
Intercept (Unacceptable)	-1.225	0.248	24.464	0.000*		
AI fear level						
low	0.613	0.226	7.346	0.007*	1.847	1.185–2.878
moderate	0.116	0.249	0.219	0.640	1.123	0.690–1.829
high	0b					
AI fear level						
low	0.230	0.527	0.190	0.663	1.258	0.448–3.535
moderate	-0.771	0.291	6.991	0.008*	0.463	0.261–0.819
high	0b					
AI harm predictability						
low	2.605	0.250	108.695	0.000*	13.529	8.291–22.077
moderate	1.365	0.261	27.298	0.000*	3.917	2.347–6.537
high	0b					
Major						
Medical	0.549	0.288	3.642	0.056	1.731	0.985–3.042
Others	0b					
Experience						
Yes	0.149	0.228	0.425	0.514	1.160	0.742–1.813
No	0b					

Abbreviations: N 990, β regression coefficient, SE standard error, p significance level, OR odds ratio, CI confidence interval, $p < .05$ indicates statistical significance

Table 8 Multinomial logistic regression predicting acceptance of AI resurrection under parental consent ($N=990$)

	β	SE	Wald	(P)	OR	95% CI
Intercept (Unacceptable)	-2.449	0.284	74.265	0.000*		
AI engagement level						
low	0.734	0.225	10.663	0.001*	2.084	1.341–3.238
moderate	0.301	0.259	1.358	0.244	1.352	0.814–2.245
high	0b					
AI fear level						
low	1.590	0.460	11.949	0.001*	4.902	1.990–12.071
moderate	0.645	0.298	4.678	0.031*	1.906	1.062–3.421
high	0b					
AI harm predictability						
low	2.095	0.239	76.790	0.000*	8.126	5.086–12.983
moderate	0.782	0.291	7.216	0.007*	2.185	1.235–3.866
high	0b					
Background						
Death is the end	0.105	0.241	0.190	0.663	1.111	0.692–1.783
Afterlife	0b					

Abbreviations: N 990, β regression coefficient, SE standard error, p significance level, OR odds ratio, CI confidence interval, $p < .05$ indicates statistical significance

correlated with acceptance across scenarios. AI fear showed scenario-dependent associations, with differing directions depending on consent conditions.

Multinomial logistic regression analysis

Five multinomial logistic regression models were estimated for the five scenarios (Tables 7, 8, 9, 10 and 11). Across scenarios, higher AI engagement and greater perceived AI harm predictability were consistently associated with higher acceptance of AI resurrection. In contrast, AI fear showed scenario-dependent associations,

with different patterns observed under explicit versus unclear consent conditions.

Discussion

The present study aimed to explore the psychological and contextual factors influencing Chinese college students' acceptance of AI resurrection across five hypothetical scenarios. This study should be interpreted in light of the sample characteristics. Participants were predominantly young adults (17–29 years) and over half were medical students, which may limit generalizability beyond highly educated Chinese university populations. Accordingly,

Table 9 Multinomial logistic regression predicting acceptance of AI resurrection under the deceased's consent ($N=990$)

	β	SE	Wald	(P)	OR	95% CI
Intercept (Unacceptable)	-2.729	0.243	126.636	0.000*		
AI engagement level						
low	0.689	0.237	8.469	0.004*	1.993	1.252–3.170
moderate	0.532	0.269	3.922	0.048*	1.703	1.006–2.885
high	0b					
AI fear level						
low	1.805	0.439	16.887	0.000*	6.083	2.571–14.391
moderate	0.282	0.331	0.728	0.394	1.326	0.693–2.536
high	0b					
AI harm predictability						
low	1.923	0.257	56.196	0.000*	6.844	4.139–11.317
moderate	1.103	0.305	13.063	0.000*	3.013	1.657–5.481
high	0b					

Abbreviations: N 990, β regression coefficient, SE standard error, p significance level, OR odds ratio, CI confidence interval, $p < .05$ indicates statistical significance

Table 10 Multinomial logistic regression predicting acceptance of AI resurrection from self-perspective ($N=990$)

	β	SE	Wald	(P)	OR	95% CI
Intercept (Disagree)	-1.203	0.393	9.350	0.002*		
AI engagement level						
low	0.734	0.204	12.970	0.000*	2.084	1.398–3.109
moderate	0.177	0.232	0.583	0.445	1.194	0.758–1.882
high	0b					
AI harm predictability						
low	2.201	0.211	108.980	0.000*	9.031	5.975–13.652
moderate	0.921	0.227	16.453	0.000*	2.512	1.610–3.920
high	0b					
Major						
	0.376	0.436	0.744	0.388	1.457	0.620–3.426
	0b					
Education						
	-0.597	0.504	1.404	0.236	0.550	0.205–1.478
	-0.215	0.378	0.322	0.571	0.807	0.384–1.694
	0b					

Abbreviations: N 990, β regression coefficient, SE standard error, p significance level, OR odds ratio, CI confidence interval, $p < .05$ indicates statistical significance

Table 11 Multinomial logistic regression predicting acceptance of AI resurrection from parent-perspective ($N=990$)

	β	SE	Wald	(P)	OR	95% CI
Intercept (Disagree)	-1.186	0.163	52.890	0.000*		
AI engagement level						
low	0.778	0.210	13.735	0.000*	2.178	1.443–3.288
moderate	0.435	0.232	3.515	0.061	1.544	0.980–2.432
high	0b					
AI fear level						
low	0.582	0.476	1.499	0.221	1.790	0.705–4.549
moderate	-0.485	0.293	2.739	0.098	0.616	0.347–1.094
high	0b					
AI harm predictability						
low	2.224	0.216	106.408	0.000*	9.241	6.056–14.099
moderate	0.962	0.237	16.465	0.000*	2.617	1.644–4.166
high	0b					

Abbreviations: N 990, β regression coefficient, SE standard error, p significance level, OR odds ratio, CI confidence interval, $p < .05$ indicates statistical significance

the findings are best understood as reflecting acceptance patterns within this subgroup rather than Chinese society as a whole.

Overall, the results indicate relatively high acceptance of AI resurrection within this student sample, with AI-related factors playing a central role in shaping acceptance. Higher levels of AI engagement were associated with greater acceptance, whereas lower perceived predictability of AI-related harm was associated with reduced acceptance. Importantly, the influence of AI fear was not uniform but varied across consent conditions, suggesting that emotional responses to AI resurrection are context-dependent. In addition, medical students and participants who had experienced the death of a loved one tended to report higher acceptance. Together, these findings highlight the central role of consent-related conditions in shaping ethical evaluations of AI resurrection.

Acceptance patterns across self- and parent-perspective scenarios

In this sample, a majority of participants did not oppose the idea of being resurrected through AI (69.1%). This proportion appears higher than that reported in earlier vignette-based work [30], although direct comparisons should be made cautiously given differences in samples and scenario framing. However, when students imagined themselves as parents deciding whether to resurrect their child, disagreement increased slightly, suggesting that attitudes vary depending on emotional involvement and moral standpoint. Such variation highlights the importance of clarifying ethical and legal responsibilities in cases where consent is uncertain or unavailable.

In hypothetical cases where an individual dies unexpectedly without surviving relatives, questions arise regarding who holds the authority to decide whether digital resurrection should occur. Such ethical dilemmas underscore the necessity of establishing clear consent procedures and robust data governance frameworks to prevent posthumous digital reanimation without explicit authorization.

AI engagement and acceptance

Across all scenarios, AI engagement was positively associated with acceptance of AI resurrection. Students who reported greater familiarity with and more frequent use of AI systems tended to evaluate AI resurrection more favourably. This pattern aligns with prior research indicating that greater exposure to AI is related to higher technological readiness and more favourable evaluations of emerging technologies [23]. In ethically sensitive applications such as AI resurrection, higher levels of engagement may be associated with reduced uncertainty and greater psychological comfort when imagining interaction with a posthumous digital entity. These findings

suggest that AI literacy efforts should not focus solely on technical competence, but also address how such systems function and what their ethical boundaries are in death-related contexts, including how consent and data governance are operationalized in posthumous contexts.

Perceived harm predictability and acceptance

Perceived harm predictability showed a consistent association with acceptance of AI resurrection across scenarios. Participants who perceived AI behaviour as unpredictable or uncontrollable tended to express lower acceptance, underscoring the importance of trust-related appraisals when AI operates in morally sensitive domains. This finding reinforces the notion that perceived transparency and controllability are critical for building trust in AI [24, 25]. In line with Human–AI trust models, transparency in algorithmic processes enables users to form cognitive trust by understanding how decisions are made [13]. Ensuring that AI resurrection systems operate in a transparent and predictable manner will be crucial for promoting both psychological comfort and social acceptance.

From a psychological perspective, this finding can be interpreted through the lens of trust-calibration theory and risk perception models. When individuals perceive AI systems as unpredictable, their sense of control and cognitive safety decreases, which may be associated with avoidance tendencies. According to dual-process theories of risk perception [36], emotional intuition often overrides rational evaluation under uncertainty, thereby relating to lower acceptance even when potential benefits are acknowledged. In the context of AI resurrection, unpredictability not only threatens technological trust but also amplifies existential concerns, as individuals may fear that the recreated entity could act beyond moral or emotional boundaries.

AI fear level shows opposite effects in different scenarios

AI fear demonstrated contrasting associations with acceptance depending on the consent context. When consent was absent or unclear, higher levels of AI fear were associated with lower acceptance. This pattern is consistent with prior findings suggesting that negative affect toward AI is linked to avoidance-oriented evaluations under conditions of uncertainty [10, 37]. In contexts where the deceased's wishes are ambiguous, AI fear may be associated with heightened moral uncertainty and existential unease, reflecting an inhibiting affective response.

Conversely, under explicit consent conditions, the direction of association appeared to reverse, with higher AI fear aligning with higher acceptance. These interpretations are necessarily post-hoc, given the exploratory design of the study. One possible explanation is that

explicit consent reduces moral ambiguity and redistributes perceived responsibility, thereby enhancing individuals' sense of legitimacy and control. Prior research in decision-making has shown that individuals adopt different cognitive and moral reasoning strategies depending on their role [38, 39]. Participants may have felt less personal control when consent had already been given, which may be associated with ambivalence toward the act itself. Under such conditions, fear may heighten awareness of the ethical gravity of AI resurrection without triggering avoidance, instead motivating engagement within a morally sanctioned framework. These findings suggest that acceptance of AI resurrection is not only determined by technology-related factors but also by context-dependent emotional and moral processes.

Importantly, the AI fear measure used in this study may capture multiple latent concerns rather than a single, uniform construct. In addition to general anxiety about AI technologies, such as fears of misuse or loss of control, AI fear in the context of resurrection may also reflect moral and existential unease related to posthumous identity, dignity, and the boundaries between life and death. These different dimensions of fear are likely to be differentially activated depending on consent clarity, which may help explain the observed scenario-dependent effects. Future research should disentangle these components using multi-item measures that distinguish technological anxiety from existential or moral threat.

Taken together, these findings suggest that the influence of AI fear on acceptance is not uniform but contingent on the consent context, supporting the role of consent scenario as a key boundary condition in evaluations of AI resurrection.

From a theoretical perspective, this pattern is broadly compatible with psychological accounts emphasizing mortality salience and existential anxiety, such as those proposed in Terror Management Theory [40]. Rather than serving as a primary explanatory framework, these perspectives provide a contextual lens for understanding why emotional responses to death-related technologies may be buffered or intensified depending on perceived moral authorization and control. Accordingly, AI fear should not be conceptualized as a uniformly inhibiting factor, as often assumed in conventional technology acceptance research, but as a context-sensitive affective response whose psychological meaning varies with consent conditions.

Although these associations were statistically reliable, most effects were small to moderate in magnitude, suggesting that acceptance of AI resurrection is likely shaped by multiple additional moral, contextual, and situational factors beyond the predictors measured in the present study.

Other factors related to students' acceptance

Disciplinary background and life experience were also associated with acceptance. Medical students demonstrated higher acceptance than non-medical students, which may be associated with their greater exposure to mortality and ethical reflection through academic and clinical training [41]. Similarly, students who had experienced the death of a loved one were more accepting of AI resurrection. This pattern echoes earlier work suggesting that bereavement increases sensitivity to death-related issues and may heighten motivation to reconnect symbolically with the deceased [42].

The deceased's consent is indispensable for individuals' AI resurrection

The deceased's consent emerged as a decisive ethical factor. Nearly half of the participants (47.9%) believed that decisions about AI resurrection should balance the wishes of the deceased and their family, while 43.9% believed the deceased's pre-mortem consent should take full precedence. Only 6.3% supported decisions made solely by the family. Acceptance tended to be higher in scenarios involving explicit consent from the deceased than in those relying on parental consent.

These findings support Harbinja's [5] argument that explicit lifetime consent is a necessary condition for digital replication or digital cloning. Future ethical and legal frameworks should prioritize autonomy, informed consent, and the dignity of the deceased when addressing AI-mediated posthumous technologies.

Cultural specificity and cross-cultural generalizability

The present findings should be interpreted within their cultural context. In China, Confucian-influenced relational perspectives on personhood emphasize family responsibility, interdependence, and moral continuity after death, which may shape attitudes toward posthumous technologies such as AI resurrection [43]. These cultural values may influence how participants interpret consent, dignity, and appropriate boundaries between the living and the deceased.

At the same time, several core psychological mechanisms identified in this study, such as trust in AI systems, affective responses to AI, and consent-based moral reasoning, are not unique to the Chinese context and may be relevant across cultures. What may differ cross-culturally is the relative salience and expression of these mechanisms rather than their fundamental presence. Future cross-cultural research is therefore needed to examine the generalizability of these findings and to clarify how cultural frameworks interact with psychological processes in shaping acceptance of AI-mediated posthumous technologies.

Theoretical implications for technology acceptance and human–AI interaction

With respect to technology acceptance theory, the present findings indicate that applying the Technology Acceptance Model (TAM) to existentially sensitive applications such as AI resurrection requires reconsideration of its core assumptions. TAM-based research typically conceptualizes acceptance as an instrumental evaluation driven by perceived usefulness and ease of use. In contrast, the findings of this study suggest that, in posthumous AI contexts, acceptance judgments are fundamentally shaped by moral legitimacy and the clarity of consent, in addition to cognitive evaluations. Rather than rejecting TAM, these results highlight its contextual limitations and suggest that cognitive assessments operate alongside ethical and affective considerations when AI technologies engage with death and identity.

Beyond TAM, this study extends understanding of human–AI interaction by demonstrating that trust, fear, and consent function differently in existential contexts than in conventional AI applications. Trust in AI in the context of resurrection is not limited to technical predictability or controllability, but also involves perceived moral alignment with human values when AI systems represent the deceased. AI fear does not act as a uniformly inhibiting factor; instead, its influence varies depending on consent conditions, reflecting a context-sensitive affective response. Consent, which is often treated as a procedural prerequisite in conventional AI systems, emerges in this study as a central factor shaping both moral reasoning and emotional responses across scenarios. Together, these findings clarify how human–AI interaction in existential domains departs from conventional AI use cases and underscore the need for context-aware approaches to technology acceptance.

Scenario-level synthesis

Across the five consent scenarios, a consistent pattern emerged in which acceptance was lowest when consent was unclear, and higher when explicit consent, either from the individual prior to death or from close family members, was present. Moreover, differences between self- and parent-perspective scenarios suggest that acceptance judgments are shaped not only by consent status but also by role-based moral reasoning, reflecting how individuals negotiate responsibility, authority, and emotional proximity in decisions about posthumous digital representation.

Limitations

The present study has several limitations that should be considered when interpreting the findings.

First, the cross-sectional design of this study precludes causal inference. Although significant associations were

identified between cognitive, affective, and demographic variables and acceptance of AI resurrection, the directionality and causality of these relationships cannot be determined from the present data. All findings should therefore be interpreted as reflecting correlational patterns rather than causal effects. Longitudinal or experimental designs are needed in future research to establish temporal precedence and clarify the causal mechanisms underlying acceptance of AI resurrection technologies.

Second, the sample was relatively homogeneous, with the majority of participants aged between 17 and 29 years and a high proportion of medical students. In addition, the sample exhibited limited variability in age and educational background. These characteristics may restrict the generalizability of the findings to broader populations, including Chinese society as a whole. Future studies should recruit more diverse samples across age groups, educational backgrounds, and social contexts to enhance external validity.

Third, AI harm predictability and AI fear were assessed using single-item measures. Although single-item measures can be appropriate in exploratory research with large samples, they may limit construct precision and reliability. Future research should employ validated multi-item scales to more comprehensively capture different dimensions of AI-related concerns, including distinctions between general technological anxiety and moral or existential unease related to posthumous technologies.

Fourth, several potentially relevant psychological variables were not measured in the present study, such as death anxiety and prior exposure to AI resurrection technologies or related narratives. These factors may influence individuals' emotional responses and ethical judgments regarding AI resurrection. Future research should incorporate these variables to further clarify the psychological mechanisms underlying acceptance and resistance toward AI-mediated posthumous technologies.

Implications

The findings of this study have important implications for education, policy, and future research on ethically sensitive AI applications, particularly those involving death, identity, and posthumous data.

Educational implications

The present findings suggest that AI literacy in ethically sensitive domains should extend beyond technical competence to include ethical reasoning, consent awareness, and cultural understanding. While prior work often frames AI literacy in terms of functional skills, our results indicate that acceptance of AI resurrection is closely tied

to individuals' cognitive familiarity with AI as well as their ethical and emotional evaluations.

However, prior research has shown that most college students do not receive structured AI instruction [22], and a U.S. survey reported that about half of students were concerned about AI's influence on education, while 60% indicated that their institutions had not clearly defined ethical guidelines for AI use [44]. Although these figures come from a U.S. context, they illustrate that ethical guidance often lags behind rapid adoption of generative AI in higher education. Accordingly, educational programs should emphasize three specific components of AI literacy that are particularly relevant for death-related AI applications.

First, ethical awareness should be strengthened by engaging students in structured discussions about moral dilemmas raised by posthumous technologies, including autonomy, dignity, identity continuity, and psychological harm. Scenario-based learning, similar to the vignettes used in this study, may be especially effective for fostering ethical reflection in emotionally complex contexts.

Second, informed consent literacy should be explicitly addressed. Students should be trained to understand consent not only as a legal requirement but as an ongoing ethical process, particularly when AI systems operate beyond the lifespan of the data subject. This includes recognizing the limits of proxy decision-making, the role of prior expressed wishes, and the long-term implications of posthumous data use.

Third, cultural sensitivity should be incorporated into AI education. Given that attitudes toward death, personhood, and family responsibility vary across cultural contexts, AI literacy programs should help students recognize how cultural frameworks shape ethical judgments. This is particularly important for AI systems intended for global deployment, where culturally insensitive design may undermine public trust and acceptance.

Ethical and policy implications

From a policy perspective, the findings underscore the need for context-specific consent frameworks for ethically sensitive AI applications such as AI resurrection. Unlike conventional AI systems, posthumous technologies raise unique challenges regarding autonomy, moral responsibility, and the authority to decide on behalf of the deceased.

Many participants favored prioritizing the deceased's wishes or balancing them with family involvement, underscoring both strong support for lifetime consent and the perceived ethical relevance of relational decision-making. This pattern highlights both strong support for lifetime consent and the perceived ethical relevance of family involvement in posthumous decision-making.

Taken together, these findings suggest that explicit lifetime consent should be treated as a primary ethical safeguard for AI resurrection, while also acknowledging the relational dimensions of decision-making in death-related contexts. Regulatory frameworks should therefore require clear, informed, and revocable consent mechanisms established during an individual's lifetime, specifying the scope, duration, and permissible uses of posthumous digital representations.

At the same time, consent frameworks should remain flexible enough to account for cultural and relational differences. In societies where family responsibility plays a central moral role, policy designs may need to balance individual autonomy with family participation, while still prioritizing the expressed wishes of the deceased. Avoiding one-size-fits-all consent models and adopting culturally adaptive governance approaches may help reduce ethical conflict, perceived uncertainty, and public resistance to posthumous AI technologies.

Research implications

This study also points to several directions for future research. First, given the exploratory nature of the present work, future studies should employ longitudinal, qualitative, or mixed-methods designs to examine how attitudes toward AI resurrection evolve over time and across different life stages or experiences with loss.

Second, future research should refine the measurement of AI-related affect by distinguishing between different dimensions of AI fear, such as technological anxiety, moral unease, and existential concern related to identity and dignity. Using validated multi-item scales would allow for a more precise examination of how emotional responses interact with trust and consent judgments in ethically sensitive AI contexts.

Finally, as suggested by Nawrat [45], advancing AI research can enhance our understanding of how to design, validate, and integrate socially responsible AI solutions. This includes improving consent frameworks, ensuring transparency, and addressing ethical concerns in AI applications, particularly those involving human identity and posthumous technologies. Comparative cross-cultural research is also needed to investigate how consent preferences, trust calibration, and affective responses to AI resurrection vary across societies with different cultural views on death, autonomy, and family responsibility. Such work would help clarify which psychological mechanisms identified in this study are culturally specific and which may generalize across contexts, thereby informing the responsible global governance of posthumous AI technologies.

Conclusions

This study contributes empirical evidence on the psychological mechanisms and contextual factors shaping acceptance of AI resurrection among Chinese college students. By integrating cognitive (AI engagement, harm predictability) and affective (AI fear) dimensions within a moral framework, the study highlights the interplay between technological trust and ethical awareness. The results emphasize that consent, transparency, and familiarity are key correlates of acceptance within this sample of public acceptance. Future research should further explore emotional and cultural factors through qualitative or cross-cultural approaches to support the ethical governance of AI resurrection and related technologies.

Abbreviations

AI	Artificial intelligence
TAM	Technology acceptance model
SD	Standard deviation
OR	Odds ratio
CI	Confidence interval
SPSS	Statistical package for the social sciences

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40359-026-04496-4>.

Supplementary Material 1.

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Authors' contributions

NC (Nini Cheng) contributed to study design, data collection, statistical analysis, and manuscript drafting. WH (Weitao Huang) supervised the project, contributed to conceptualization, methodology, and data interpretation, and critically revised the manuscript. MS (Mei Sun) provided overall supervision, contributed to study design, and reviewed and approved the final version of the manuscript. All authors read and approved the final manuscript.

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Data availability

The datasets generated and/or analysed during the current study are not publicly available due to ethical and privacy restrictions but are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was conducted in accordance with the ethical principles of the Declaration of Helsinki. Ethical approval was obtained from the Ethics Committee of Xiangya School of Nursing, Central South University. All participants provided electronic informed consent prior to participation. Respondents were informed of the study's purpose, voluntary nature, confidentiality assurances, and their right to withdraw at any time without consequence.

Consent for publication

Not applicable. This manuscript does not contain any individual person's data in any form (including individual details, images, or videos).

Competing interests

The authors declare no competing interests.

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